QF600 Asset Pricing — Week 6 Reference Cheatsheet

Stochastic Discount Factor: consumption–investment choice, the Euler equation, E[M̃R̃ᵢ] = 1, the pricing kernel

Built from Lecture 6 slides (Liu Peng, LKCSB) · September 2026

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*Source: "Lecture slides – Week 6 v2" (Liu Peng, LKCSB, 17 slides). Slide numbers are PDF page numbers and are shown as [S#]. Material not on the slides is tagged **[Beyond slides]**. ⚠ marks slide ambiguities or errors, with the corrected version.*

*Exam context (Week 1 [S5]): the final is closed book, with one double-sided A4 sheet allowed. It is 40% MCQ and 60% written (concepts plus simple calculations), with **no coding**. For this week you need: the consumption–portfolio problem, both first-order conditions and how they combine into $E[\tilde{M}\tilde{R}_i] = 1$, the definition $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$, the price form $P_i = E[\tilde{M}\tilde{X}_i]$, and the "large in bad states" logic. The [Beyond slides] consequences (risk-free rate, covariance*

form, state prices, CAPM as a special case) are the standard next steps and are cheap to learn from here.

⚠ **Slide cross-references.** The deck refers to its own slides by an internal number that is **3 lower** than the PDF page: "slide 9" = [S12], "slide 10" = [S13], "slide 11" = [S14]. This sheet always uses PDF pages.

Notation used throughout (slide conventions, [S7]–[S8])

Symbol	Meaning	Dim / units	Time	Notes
W_0	wealth today	\$	date 0	Known
C_0	consumption today (the choice)	\$	date 0	Known once chosen
$S = W_0 - C_0$	amount saved / invested	\$	date 0	Known
w_i	weight of asset i in the invested pot	fraction	date 0	$\sum_i w_i = 1$
$\tilde{R}_i$	**gross** return on asset i (what \$1 becomes)	ratio, e.g. 1.08	date 1	Random except the riskless asset [S8]
$\tilde{R}_p = \sum_i w_i \tilde{R}_i$	portfolio gross return	ratio	date 1	
$\tilde{W}_1 = \tilde{C}_1$	date-1 wealth, all consumed	\$	date 1	$= (W_0 - C_0)\tilde{R}_p$
$U(\cdot)$	utility of consumption (same function both dates)	utils	—	$U' > 0$, $U'' < 0$ [S6]
δ	time-preference ("patience") factor	scalar, usually < 1	—	Multiplies tomorrow's expected utility
$V(C_0, \tilde{C}_1)$	lifetime (time-separable) utility	utils	—	$U(C_0) + \delta E[U(\tilde{C}_1)]$
λ	Lagrange multiplier on $\sum w_i = 1$	utils per unit weight	—	⚠ Not risk aversion, not W5's factor premium. Equals $(W_0 - C_0)U'(C_0)$ [Beyond slides]
$\tilde{M}$	stochastic discount factor $\delta U'(\tilde{C}_1)/U'(C_0)$	per \$ (dimensionless)	date 1	a.k.a. intertemporal MRS, pricing kernel [S15]
$\tilde{X}_i$	payoff of asset i at date 1	\$	date 1	
P_i	price of asset i today	\$	date 0	$\tilde{R}_i = \tilde{X}_i / P_i$
*	evaluated at the optimum	—	—	e.g. C_0^*, w_i^*
R_f	gross risk-free return [Beyond slides]	ratio	known at 0	$= 1/E[\tilde{M}]$
$s,\ p_s,\ q_s$	state, its probability, its state price [Beyond slides]	—	—	$q_s = p_s M_s$

Tilde = random [S4]: known amount × unknown return = unknown, so $\tilde{C}_1$ carries a tilde and S does not.

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0. Topic map

```
PREREQUISITES
 W1 Expected utility E[U(W)], U' > 0, U'' < 0 (risk aversion), certainty equivalents
 W2 Portfolio returns R_p = Σ w_i R_i, Σ w_i = 1
 W3 CAPM: μ_i − R_f = β_i(μ_M − R_f); negative β ⇒ μ < R_f ("insurance")
 W4 Multi-factor models: premia for exposure to factors
 W5 Factor-neutral books give up the factor premium λ_f
        │
        ▼
WEEK 6: THE STOCHASTIC DISCOUNT FACTOR
 ├─ SETUP [S2–S6]
 │   ├─ Spend or invest: C0 + S = W0  [S2–S3]
 │   ├─ Uncertainty: C̃1 = (W0 − C0) R̃  [S4]
 │   └─ Diminishing marginal utility: U' > 0, U'' < 0 ⇒ smooth consumption  [S5–S6]
 │
 ├─ ACT 1  The consumption–portfolio problem  [S7–S14]
 │   ├─ One period, two dates; two decisions (consume/save, how to invest)  [S7]
 │   ├─ Budget: Σ w_i = 1,  C̃1 = (W0 − C0) Σ w_i R̃_i  [S8]
 │   ├─ Objective: V = U(C0) + δ E[U(C̃1)]  (time-separable)  [S9–S10]
 │   ├─ Problem + Lagrangian; two FOCs  [S11]
 │   ├─ FOC in C0 → Euler:  U'(C0*) = δ E[U'(C̃1*) Σ w_i* R̃_i]  [S12]
 │   ├─ FOC in w_i → allocation:  E[U'(C̃1*) R̃_i] equal for all i  [S13]
 │   └─ Combine: E[ δU'(C̃1)/U'(C0) · R̃_i ] = 1 for every asset  [S14]
 │
 ├─ ACT 2  The SDF  [S15–S16]
 │   ├─ M̃ = δ U'(C̃1)/U'(C0): IMRS = SDF = pricing kernel  [S15]
 │   ├─ E[M̃ R̃_i] = 1  ⇔  P_i = E[M̃ X̃_i]  [S15]
 │   └─ M̃ is largest in bad states ⇒ insurance dear (low E[R]), boom assets cheap  [S16]
 │
 ├─ CONSEQUENCES  [Beyond slides]
 │   ├─ R_f = 1/E[M̃];  E[R̃_i] − R_f = −R_f Cov(M̃, R̃_i)
 │   ├─ P = E[X̃]/R_f + Cov(M̃, X̃);  state prices q_s = p_s M_s
 │   ├─ Hansen–Jagannathan bound: |Sharpe| ≤ σ(M̃)/E[M̃]
 │   └─ M̃ linear in R̃_M ⇒ CAPM;  linear in factors ⇒ W4 factor model
 │
 └─ In-class quiz Q1–3 [S17] (title only)
        │
        ▼
 NEXT: every pricing model = a choice of M̃ (CAPM, multi-factor, consumption-based, risk-neutral pricing)
```

Dependency logic:

- The **setup** gives two ingredients: a budget that links today to tomorrow, and a concave U that makes a dollar more valuable when consumption is low.
- **Act 1** optimises. One FOC says "eat or invest the marginal dollar, you can't gain either way"; the other says "which asset, every asset gives the same marginal-utility-weighted return".
- **Act 2** divides by $U'(C_0)$, names the ratio $\tilde{M}$, and rewrites returns as prices. Everything that follows (and every model from W3–W5) is a special case of $P = E[\tilde{M}\tilde{X}]$.

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1. Concept entries

1.1 Spend or invest: the one trade-off [S2–S3]

- **Question [S2]**: "How do you spend your salary?"
- **Statement [S3]**: every dollar earned today is either consumed or invested:

$$C_0 + S = W_0, \qquad S = W_0 - C_0.$$

  - C_0: "SPEND — consume it now".
  - S: "INVEST — set it aside to grow".
- **Intuition**: every dollar spent today is a dollar that can't be invested. Investing more trades enjoyment now for more money later. You choose the split.
- **Traps**:
  - S is a **level** (\$), not a rate. The saving *rate* would be S/W_0.
  - There is no labour income at date 1 in this model. Date-1 consumption comes only from savings [Beyond slides: adding income Y_1 just gives $\tilde{C}_1 = S\tilde{R}_p + Y_1$; the SDF definition is unchanged].

1.2 Uncertainty and the tilde [S4]

- **Statement**:

$$\underbrace{S = W_0 - C_0}_{\text{known today}} \times \underbrace{\tilde{R}}_{\text{random}} = \underbrace{\tilde{C}_1}_{\text{random}} .$$

  - Example on the slide: \$1 could become \$0.90, \$1.00 or \$1.20; you don't know yet.
- **Key convention [S8]**: $\tilde{R}$ is the **gross** return, "what \$1 becomes". A 5% gain is $\tilde{R} = 1.05$, not 0.05.
- **Traps**:
  - Using net returns r in $E[\tilde{M}\tilde{R}] = 1$. With net returns the equation is $E[\tilde{M}(1+\tilde{r})] = 1$, i.e. $E[\tilde{M}\tilde{r}] = 1 - E[\tilde{M}]$ [Beyond slides]. Plugging net returns into "=1" is wrong.
  - Only S is chosen; the distribution of $\tilde{R}$ is given by the market.

1.3 Diminishing marginal utility [S5–S6]

- **Statement [S5]**: consuming one dollar is worth more when you are poor.
  - ⚠ The slide reads "Consuming 1 dollar is worth more you are poor"; the word **"when"** is missing.
- **Properties [S6]**:
  - $U'(C) > 0$: the curve slopes up; more is better.
  - $U''(C) < 0$: the curve bends down; each extra unit adds less.
  - Picture: the same +\$1 gives a "big jump in well-being" at low consumption and "barely a nudge" at high consumption.
- **Implication [S6]**: "the smart move is to smooth consumption over time". Because U is concave, moving a dollar from a rich date to a poor date raises total utility.
- **Common utility functions [Beyond slides]**:

$U(C)$	$U'(C)$	Relative risk aversion $-CU''/U'$	SDF $\tilde{M}$
$\ln C$	$1/C$	1	$\delta\, C_0/\tilde{C}_1$
$\frac{C^{1-\gamma}}{1-\gamma}$ (CRRA / power)	$C^{-\gamma}$	γ	$\delta\,(\tilde{C}_1/C_0)^{-\gamma}$
$C - \frac{b}{2}C^2$ (quadratic)	$1 - bC$	$bC/(1-bC)$	$\delta\,\frac{1-b\tilde{C}_1}{1-bC_0}$ — **linear** in $\tilde{C}_1$

- **Interpretation**: $U'' < 0$ is the same property that made investors risk averse in W1. Here it does double duty: it makes people dislike **risk** (at one date) and dislike **uneven consumption** (across dates and states).
- **Trap**: marginal utility U' (a slope) is what prices assets, not the level U.
- **Connections**: W1 (concavity ⇒ risk aversion, Jensen's inequality $E[U(\tilde{C})] < U(E[\tilde{C}])$).

1.4 One period, two dates: the budget [S7–S8]

- **Timeline [S7]**:
  - Date 0: start with W_0. **Decision 1** (consume or save?): consume C_0, invest $S = W_0 - C_0$.
  - **Decision 2** (how to invest?): split S across assets, earning $\tilde{R}_p = \sum_i w_i \tilde{R}_i$.
  - Date 1: all wealth is consumed, $\tilde{C}_1 = \tilde{W}_1$.
- **Two budget conditions [S8]**:

$$\text{(1) everything is invested: } \sum_i w_i = 1, \qquad \text{(2) } \tilde{C}_1 = (W_0 - C_0)\sum_i w_i \tilde{R}_i.$$

  - "The invested pot $W_0 - C_0$, split so the shares add to 100%."
  - "What you saved, grown by the portfolio return, all eaten at date 1."
- **Assumptions**:
  - One period; no bequest (date-1 wealth is fully consumed).
  - w_i can be any real number (shorting allowed); only the budget constraint binds. This is what makes the FOCs hold with **equality** for every asset [Beyond slides: the deck never states it, but the derivations on [S12]–[S14] need it].
  - The riskless asset, if present, is one of the i's (its $\tilde{R}_i$ is not random) [S8].
- **Traps**:
  - w_i are weights **of the invested pot**, not of total wealth W_0.
  - With short-sale constraints, an asset held at a corner ($w_i = 0$) need not satisfy the FOC with equality [Beyond slides]; then $E[\tilde{M}\tilde{R}_i] \leq 1$ for that asset.

1.5 The objective: time-separable expected utility [S9–S10]

- **Statement**:

$$V(C_0, \tilde{C}_1) = U(C_0) + \delta\, E[U(\tilde{C}_1)].$$

  - Today: enjoy consumption now, $U(C_0)$.
  - Tomorrow (uncertain): the expected utility of what savings buy, $E[U(\tilde{C}_1)]$.
  - δ ("patience") discounts tomorrow's utility. $\delta < 1$ shrinks it.
  - "One number to rank any plan": choose C_0 and weights w_i to make it as large as possible [S9].
- **Picture [S10]**: each date's consumption passes through **one** concave curve; V stacks $U(C_0)$ and $\delta E[U]$.
- **Time-separable [S10]**: today's satisfaction and tomorrow's don't interact. There is no $\partial^2 V / \partial C_0 \partial C_1$ term (no habits, no durables).
  - "A strong assumption, and one that later models relax" (e.g. habit formation, Epstein–Zin recursive utility [Beyond slides]).
- **Two separate roles [Beyond slides]**:
  - δ = impatience (the pure time preference). A higher δ = more patient = save more.
  - The curvature of U = both risk aversion and the dislike of uneven consumption over time. With power utility, the elasticity of intertemporal substitution is $1/\gamma$, tied to risk aversion γ. Epstein–Zin separates them.
- **Traps**:
  - δ discounts **utility**, not dollars. The discount on dollars is $\tilde{M}$, which also contains $U'(\tilde{C}_1)/U'(C_0)$.
  - $E[U(\tilde{C}_1)] \neq U(E[\tilde{C}_1])$ (Jensen). Expected utility, not utility of expected consumption.

1.6 The consumption and portfolio choice problem [S11]

- **Problem**:

$$\max_{C_0,\, w} \; U(C_0) + \delta\, E[U(\tilde{C}_1)] \quad \text{s.t.} \quad \sum_i w_i = 1, \qquad \tilde{C}_1 = (W_0 - C_0) \sum_i w_i \tilde{R}_i.$$

- **Three steps [S11]**:
  1. Form the Lagrangian: $\mathcal{L} = U(C_0) + \delta E[U(\tilde{C}_1)] + \lambda(1 - \sum_i w_i)$.
  2. Differentiate in C_0: gives the **Euler equation** (1.7).
  3. Differentiate in each w_i: gives the **allocation rule** (1.8).
- "Two first-order conditions, and between them they contain the whole of modern asset pricing."
- ⚠ **Sign convention**: [S11] writes $+\lambda(1 - \sum w_i)$, [S13] writes $-\lambda(\sum w_j - 1)$. These are **identical**. Also [S13] drops $U(C_0)$ from the weight problem because it doesn't depend on the split; that changes nothing in $\partial / \partial w_i$.
- **Assumptions**: interior optimum, U differentiable, expectation and derivative can be interchanged.

1.7 First condition: eat or invest? (the Euler equation) [S12]

- **Derivation (exam-reproducible)**:
  1. Differentiate V in C_0, substituting the budget so $\tilde{C}_1$ depends on C_0:
  $$\frac{dV}{dC_0} = U'(C_0) + \delta\, E\Big[U'(\tilde{C}_1)\cdot\frac{\partial \tilde{C}_1}{\partial C_0}\Big] = 0.$$
  2. A dollar not eaten is invested. Only $S = W_0 - C_0$ depends on C_0, with slope -1:
  $$\tilde{C}_1 = (W_0 - C_0)\sum_i w_i^* \tilde{R}_i \;\Rightarrow\; \frac{\partial \tilde{C}_1}{\partial C_0} = -\sum_i w_i^* \tilde{R}_i.$$
  3. Substitute and rearrange:
  $$\boxed{U'(C_0^*) = \delta\, E\Big[U'(\tilde{C}_1^*)\sum_i w_i^* \tilde{R}_i\Big]}$$
- **Reading**: left = utility lost by eating one dollar less today. Right = expected (discounted) utility gained by investing that dollar in the optimal portfolio and eating the proceeds tomorrow. At the optimum they balance: "you can't gain by moving a dollar either way".
- **Worked example [Beyond slides] — log utility**:
  - $U = \ln C$, so $U'(C) = 1/C$ and $\tilde{C}_1 = (W_0 - C_0)\tilde{R}_p$.
  - Euler: $\frac{1}{C_0} = \delta E\Big[\frac{\tilde{R}_p}{(W_0 - C_0)\tilde{R}_p}\Big] = \frac{\delta}{W_0 - C_0}$.
  - So $C_0^* = \frac{W_0}{1+\delta}$ and $S^* = \frac{\delta W_0}{1+\delta}$. **The log investor's consumption doesn't depend on returns at all** (income and substitution effects cancel).
  - Numbers: $W_0 = 196$, $\delta = 0.96$ gives $C_0^* = 100$ and $S^* = 96$. (Used again in 1.10–1.12.)
- **Traps**:
  - Forgetting the minus sign from $\partial \tilde{C}_1/\partial C_0$.
  - Pulling $U'(\tilde{C}_1)$ out of the expectation. It is random and correlated with $\tilde{R}_p$: $E[U'\tilde{R}_p] \neq E[U']E[\tilde{R}_p]$.
- **Connections**: with a riskless asset only, the Euler equation is $U'(C_0) = \delta R_f E[U'(\tilde{C}_1)]$, the textbook consumption-smoothing condition [Beyond slides].

1.8 Second condition: which asset? (the allocation rule) [S13]

- **Derivation (exam-reproducible)**:
  1. Start from the constrained choice. $U(C_0)$ doesn't depend on the split, so maximise only the future term: $\max_w\ \delta E[U(\tilde{C}_1)]$ s.t. $\sum_j w_j = 1$.
  2. Fold the constraint in: $\mathcal{L} = \delta E[U(\tilde{C}_1)] - \lambda(\sum_j w_j - 1)$.
  3. Differentiate in w_i, using $\partial \tilde{C}_1/\partial w_i = (W_0 - C_0)\tilde{R}_i$:
  $$\delta(W_0 - C_0)\, E[U'(\tilde{C}_1)\tilde{R}_i] = \lambda.$$
  4. λ does not depend on i, so every asset's marginal-utility-weighted return is the same:
  $$\boxed{E[U'(\tilde{C}_1^*)\tilde{R}_i] = E[U'(\tilde{C}_1^*)\tilde{R}_j] \quad \text{for all } i, j.}$$
- **Reading**: at the optimum, shifting a sliver of the pot from asset j to asset i doesn't change expected utility. A return counts for more when it arrives in a state where U' is high (consumption low).
- **Value of λ [Beyond slides]**: combine with 1.7 (see 1.9): $\lambda = (W_0 - C_0)\,U'(C_0^*)$. It is the utility value of the whole invested pot at today's marginal utility.
- **Traps**:
  - The equal quantity is $E[U'\tilde{R}_i]$, **not** $E[\tilde{R}_i]$. Assets have different expected returns precisely because they covary differently with U'.
  - λ here has nothing to do with the risk-aversion λ of W2/W5 MVO.

1.9 Putting it together: $E[\tilde{M}\tilde{R}_i] = 1$ for every asset [S14]

- **Derivation [S14]**:
  1. **It holds per asset, not just for the portfolio.** The Euler equation (1.7) is a weighted average of the per-asset terms; 1.8 says they're all equal; with $\sum w_i^* = 1$ the average is just that common term:

$$U'(C_0) = \delta \sum_i w_i^* E[U'(\tilde{C}_1)\tilde{R}_i] = \delta\, E[U'(\tilde{C}_1)\tilde{R}_i].$$

  2. **Divide by today's marginal utility.** $U'(C_0)$ is a known number:

$$E\Big[\frac{\delta U'(\tilde{C}_1)}{U'(C_0)}\tilde{R}_i\Big] = 1.$$

  3. **Name the ratio** $\tilde{M}$: one number per future state, the same for every asset:

$$\boxed{E[\tilde{M}\tilde{R}_i] = 1, \qquad \tilde{M} = \frac{\delta U'(\tilde{C}_1)}{U'(C_0)}.}$$

- "Every asset now obeys one clean rule."
- **Assumptions carried**: interior optimum for every asset (unconstrained weights), time-separable expected utility, gross returns.
- **Trap**: the result holds for any asset the investor **could** trade at the margin (even at $w_i = 0$), as long as no constraint binds on it. It is not a statement only about assets actually held.

1.10 The Stochastic Discount Factor [S15]

- **Definition**:

$$\tilde{M} = \delta \frac{U'(\tilde{C}_1)}{U'(C_0)}.$$

  - One random number for every future state of the world.
  - Large when a future dollar is scarce, small when plentiful. "That is the discount."
- **Three names, one object [S15]**:
  - **Intertemporal MRS**: a marginal rate of substitution is how much of one good you'd trade for another and stay equally happy. Across time, it is how many units of today's consumption a future dollar in that state is worth.
    - ⚠ The slide's box says "how many units of today's **happiness** a future dollar is worth". $\tilde{M}$ is measured in today's **dollars**, not utils: $\delta U'(\tilde{C}_1)$ is the (discounted) utility a future dollar brings, and dividing by $U'(C_0)$ (utils per dollar today) converts it into today's dollars. That is why $P = E[\tilde{M}\tilde{X}]$ comes out in \$.
  - **Stochastic discount factor**: a discount factor (like $1/(1+r)$) that is random, different in each state.
  - **Pricing kernel**: the one object that prices every asset.
- **From returns to prices [S15]**:
  1. Start from $E[\tilde{M}\tilde{R}_i] = 1$.
  2. A return is payoff ÷ price, $\tilde{R}_i = \tilde{X}_i/P_i$, so $E[\tilde{M}\tilde{X}_i/P_i] = 1$.
  3. P_i is known today; pull it out:

$$\boxed{P_i = E[\tilde{M}\tilde{X}_i].}$$

  - "Today's price = expected payoff, discounted by M. The same one line prices anything with a payoff — a stock, a bond, an option, a factory."
- **Worked example [Beyond slides]** (continuing 1.7: $U = \ln C$, $\delta = 0.96$, $C_0 = 100$):
  - Three states for $\tilde{C}_1$: recession 90 ($p = 0.3$), normal 105 ($p = 0.4$), boom 120 ($p = 0.3$).
  - $M_s = \delta C_0/C_{1,s} = 0.96 \times 100/C_{1,s}$, so $\tilde{M} = (1.0667,\ 0.9143,\ 0.8000)$ and $E[\tilde{M}] = 0.9257$.
  - Price three payoffs with $P = \sum_s p_s M_s X_s$:

Asset	Payoff (rec, norm, boom)	Price P	$E[\tilde{R}]$	Premium $E[\tilde{R}] - R_f$
Riskless bond	(1, 1, 1)	0.9257	1.0802	0
"Insurance" (pays only in recession)	(1, 0, 0)	0.3200	0.9375	**−14.3%**
"Stock"	(80, 105, 130)	95.20	1.1029	**+2.3%**

- Stock: $0.3(1.0667)(80) + 0.4(0.9143)(105) + 0.3(0.8)(130) = 25.6 + 38.4 + 31.2 = 95.2$.
- Check $E[\tilde{M}\tilde{R}] = 1$ for each (e.g. insurance: $0.3 \times 1.0667 \times \frac{1}{0.32} = 1$ ✓).
- Consistency check: the investor's own portfolio earns $\tilde{R}_p = \tilde{C}_1/S = (90, 105, 120)/96 = (0.9375,\ 1.0938,\ 1.25)$. With log utility $M_s R_{p,s} = \delta C_0/S = 0.96 \times 100/96 = 1$ in **every** state, so $E[\tilde{M}\tilde{R}_p] = 1$ ✓. The example is internally consistent with the optimum of 1.7.
- **Traps**:
  - $P = E[\tilde{M}\tilde{X}]$, not $E[\tilde{M}]E[\tilde{X}]$. The difference is the covariance (risk adjustment, 1.12).
  - $\tilde{M}$ is **the same** for all assets; only the payoffs differ.
  - Units: $\tilde{X}$ in \$, P in \$; $\tilde{M}$ is "\$ today per \$ in that state".

1.11 What the SDF does: largest when a dollar is scarce [S16]

- **Statement**: $\tilde{M}$ falls as future consumption $\tilde{C}_1$ rises, because U' falls (concave utility). The chart plots $\tilde{M}$ against $\tilde{C}_1$ with points at Recession (high), Normal, Boom (low).
- **Reading the curve [S16]**:
  - M_s is the value today of one dollar delivered in state s (per unit probability; see state prices in 1.12).
  - It is biggest in bad states: low consumption means an extra dollar is precious.
  - So $P_i = E[\tilde{M}\tilde{X}_i]$ pays the most for payoffs that arrive in bad states.
- **Conclusion [S16]**:
  - **Insurance-like assets** (pay in a slump) are **dear**, i.e. a high price and a **low expected return**.
  - **Boom assets** (pay when times are good) are **cheap**, i.e. a low price and a **high expected return**.
- **Worked example**: 1.10's table: the recession-only claim has $E[\tilde{R}] = 0.9375 < R_f$ (it *loses* 6.25% on average and is still rationally held); the pro-cyclical stock earns a premium.
- **Traps**:
  - "Risky" means covarying with consumption (with $\tilde{M}$), not having high variance. A very volatile payoff that is independent of $\tilde{M}$ earns R_f (see 1.12).
  - Direction: high $\tilde{M}$ state ⇒ payoffs there are *valuable* ⇒ price *up* ⇒ expected return *down*.
- **Connections**: exactly the W3 negative-β logic ("the payoff arrives exactly when a dollar is worth most"), now derived from preferences rather than from mean–variance equilibrium.

1.12 Consequences of $E[\tilde{M}\tilde{R}] = 1$ [Beyond slides]

The slides stop at [S16]. These five results follow in two lines each and are the standard toolkit for the rest of the course.

(a) The risk-free rate. For a riskless asset $\tilde{R} = R_f$ (a constant), so $R_f E[\tilde{M}] = 1$:

$$R_f = \frac{1}{E[\tilde{M}]}.$$

- Example 1.10: $R_f = 1/0.9257 = 1.0802$ (8.0%).
- With CRRA: a higher expected consumption growth or lower δ raises R_f (people want to borrow against a richer future).

(b) Covariance (risk-premium) form. Using $E[\tilde{M}\tilde{R}_i] = E[\tilde{M}]E[\tilde{R}_i] + \text{Cov}(\tilde{M}, \tilde{R}_i) = 1$:

$$E[\tilde{R}_i] - R_f = -R_f \,\text{Cov}(\tilde{M}, \tilde{R}_i).$$

- Positive covariance with $\tilde{M}$ (pays in bad times) ⇒ **negative** premium; negative covariance (pays in good times) ⇒ positive premium; zero covariance ⇒ $E[\tilde{R}_i] = R_f$ however volatile.
- Price version: $P_i = \dfrac{E[\tilde{X}_i]}{R_f} + \text{Cov}(\tilde{M}, \tilde{X}_i)$ = "risk-neutral PV + risk adjustment".
- Example 1.10 (stock): $E[\tilde{X}]/R_f = 105/1.0802 = 97.2$ and $\text{Cov}(\tilde{M}, \tilde{X}) = 95.2 - 0.9257 \times 105 = -2.0$, so $P = 95.2$ ✓. Premium $= -R_f \text{Cov}(\tilde{M}, \tilde{R}) = +2.27\%$ ✓.

(c) State prices and risk-neutral probabilities. With discrete states,

$$P = \sum_s p_s M_s X_s = \sum_s q_s X_s, \qquad q_s = p_s M_s = \text{price today of \$1 in state } s \text{ only.}$$

- Risk-neutral probabilities $\pi_s^Q = q_s R_f$ sum to 1, and $P = E^Q[\tilde{X}]/R_f$.
- Example 1.10: $q = (0.320, 0.366, 0.240)$ and $\pi^Q = (0.346, 0.395, 0.259)$ vs true $p = (0.3, 0.4, 0.3)$. Risk-neutral probabilities **overweight bad states**.

(d) Hansen–Jagannathan bound. From (b) and $|\text{Corr}| \le 1$:

$$\frac{|E[\tilde{R}_i] - R_f|}{\sigma(\tilde{R}_i)} \le \frac{\sigma(\tilde{M})}{E[\tilde{M}]}.$$

- Any asset's Sharpe ratio is capped by the SDF's volatility relative to its mean. High observed Sharpe ratios require a volatile $\tilde{M}$.
- Example 1.10: $\sigma(\tilde{M}) = 0.1037$, bound = 0.112; the stock's Sharpe = 0.112 (almost exactly on the bound because its payoff is almost perfectly negatively correlated with $\tilde{M}$).
- (With CRRA and ~2% consumption volatility, $\sigma(M)/E[M] \approx \gamma\sigma_{\Delta c}$, so matching an equity Sharpe of ~0.4 needs $\gamma \approx 20$: the **equity premium puzzle**, Mehra–Prescott 1985.)

(e) CAPM and factor models are special cases. Suppose $\tilde{M} = a - b\tilde{R}_M$ (linear in the market return, e.g. quadratic utility with everyone holding the market). Then from (b):

$$E[\tilde{R}_i] - R_f = R_f b \,\text{Cov}(\tilde{R}_M, \tilde{R}_i), \qquad E[\tilde{R}_M] - R_f = R_f b \,\text{Var}(\tilde{R}_M).$$

- Divide: $E[\tilde{R}_i] - R_f = \beta_i (E[\tilde{R}_M] - R_f)$ with $\beta_i = \text{Cov}/\text{Var}$. **The CAPM.**
- $b > 0$ ⇒ $\tilde{M}$ high when the market is low, which is exactly why negative-β assets earn less than R_f (W3 [S42]).
- $\tilde{M} = a - b^\top \tilde{f}$ (linear in K factors) gives the W4 multi-factor beta-pricing model.
- So "which asset pricing model?" = "which $\tilde{M}$?".

1.13 In-class quiz [S17]

- Title slide only ("Questions 1–3"); no questions appear in the deck. See §5 for practice questions.

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2. Comparison tables

2.1 The two first-order conditions

	FOC in C_0 (Euler) [S12]	FOC in w_i (allocation) [S13]
Question	eat or invest the marginal dollar?	which asset gets the marginal dollar?

	FOC in C_0 (Euler) [S12]	FOC in w_i (allocation) [S13]
Perturbation	move \$1 between today and the portfolio	move weight between assets i and j
Key derivative	$\partial \tilde{C}_1/\partial C_0 = -\sum w_i^* \tilde{R}_i$	$\partial \tilde{C}_1/\partial w_i = (W_0 - C_0)\tilde{R}_i$
Condition	$U'(C_0^*) = \delta E[U'(\tilde{C}_1^*)\tilde{R}_p]$	$\delta(W_0 - C_0)E[U'(\tilde{C}_1)\tilde{R}_i] = \lambda$, same for all i
Pins down	the level of saving	the relative pricing of assets
Combined	$E[\tilde{M}\tilde{R}_i] = 1$ for every i [S14]	

2.2 Three ways to write the same pricing statement

Form	Equation	Best for
Return form [S14]	$E[\tilde{M}\tilde{R}_i] = 1$	testing with return data
Price form [S15]	$P_i = E[\tilde{M}\tilde{X}_i]$	valuing any payoff (bond, stock, option, project)
Covariance form [Beyond]	$E[\tilde{R}_i] - R_f = -R_f \mathrm{Cov}(\tilde{M}, \tilde{R}_i)$	explaining risk premia
State-price form [Beyond]	$P_i = \sum_s q_s X_{i,s}$, $q_s = p_s M_s$	discrete-state problems
Risk-neutral form [Beyond]	$P_i = E^Q[\tilde{X}_i]/R_f$	derivatives pricing

2.3 Good-state vs bad-state payoffs [S16]

	Pays in bad states (recession)	Pays in good states (boom)	Independent of $\tilde{M}$
$\mathrm{Cov}(\tilde{M}, \tilde{R})$	> 0	< 0	0
Price	high (“dear”)	low (“cheap”)	$E[\tilde{X}]/R_f$
Expected return	$< R_f$	$> R_f$	$= R_f$
Example	insurance, index puts, (sometimes) gold, Treasuries in a crash	cyclical stocks	purely idiosyncratic bets
W3 analogue	$\beta < 0$	$\beta > 0$	$\beta = 0$

2.4 SDF vs CAPM (Week 3)

	CAPM (W3)	SDF (W6)
Starting point	mean–variance investors + market clearing	any expected-utility investor’s FOCs
What prices risk	$\beta_i = \mathrm{Cov}(R_i, R_M)/\mathrm{Var}(R_M)$	$\mathrm{Cov}(\tilde{M}, \tilde{R}_i)$
Bad state defined by	low market return	low consumption (high U')
Pricing equation	$\mu_i - R_f = \beta_i(\mu_M - R_f)$	$E[\tilde{M}\tilde{R}_i] = 1$
Relation	special case with $\tilde{M} = a - b\tilde{R}_M$	general framework
Assumptions needed	homogeneous expectations, MV preferences	time-separable EU, interior optimum

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3. Master formula sheet

Budget [S3, S7–S8]

- $C_0 + S = W_0$, $S = W_0 - C_0$.
- $\sum_i w_i = 1$; $\tilde{R}_p = \sum_i w_i \tilde{R}_i$ (gross).
- $\tilde{C}_1 = \tilde{W}_1 = (W_0 - C_0)\sum_i w_i \tilde{R}_i$.

Preferences [S6, S9–S10]

- $U' > 0, U'' < 0$.
- $V(C_0, \tilde{C}_1) = U(C_0) + \delta E[U(\tilde{C}_1)]$.

Problem [S11]

- $\max_{C_0, w} U(C_0) + \delta E[U(\tilde{C}_1)]$ s.t. $\sum_i w_i = 1$.
- $\mathcal{L} = U(C_0) + \delta E[U(\tilde{C}_1)] + \lambda(1 - \sum_i w_i)$.

FOCs [S12–S13]

- $\partial \tilde{C}_1 / \partial C_0 = -\sum_i w_i^* \tilde{R}_i$ and $\partial \tilde{C}_1 / \partial w_i = (W_0 - C_0)\tilde{R}_i$.
- Euler: $U'(C_0^*) = \delta E[U'(\tilde{C}_1^*) \sum_i w_i^* \tilde{R}_i]$.
- Allocation: $\delta(W_0 - C_0)E[U'(\tilde{C}_1)\tilde{R}_i] = \lambda \Rightarrow E[U'(\tilde{C}_1^*)\tilde{R}_i] = E[U'(\tilde{C}_1^*)\tilde{R}_j]$.
- [Beyond] $\lambda = (W_0 - C_0)U'(C_0^*)$.

SDF [S14–S16]

- $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$.
- $E[\tilde{M}\tilde{R}_i] = 1$ for every asset i.
- $\tilde{R}_i = \tilde{X}_i / P_i \Rightarrow P_i = E[\tilde{M}\tilde{X}_i]$.
- $\tilde{M}$ decreasing in $\tilde{C}_1$ (because $U'' < 0$).

Special utilities [Beyond]

- Log: $\tilde{M} = \delta C_0 / \tilde{C}_1$; $C_0^* = W_0/(1+\delta)$.
- CRRA: $\tilde{M} = \delta(\tilde{C}_1 / C_0)^{-\gamma}$.
- Quadratic: $\tilde{M}$ linear in $\tilde{C}_1$.

Consequences [Beyond]

- $R_f = 1/E[\tilde{M}]$.
- $E[\tilde{R}_i] - R_f = -R_f \mathrm{Cov}(\tilde{M}, \tilde{R}_i)$.
- $P_i = E[\tilde{X}_i]/R_f + \mathrm{Cov}(\tilde{M}, \tilde{X}_i)$.
- $q_s = p_s M_s$; $\pi_s^Q = q_s R_f$; $P = \sum_s q_s X_s = E^Q[\tilde{X}]/R_f$.
- HJ: $|E[\tilde{R}] - R_f| / \sigma(\tilde{R}) \le \sigma(\tilde{M})/E[\tilde{M}]$.
- $\tilde{M} = a - b\tilde{R}_M \Rightarrow$ CAPM; $\tilde{M} = a - b^\top \tilde{f} \Rightarrow$ multi-factor beta pricing.

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4. Glossary

- **Consumption smoothing**: spreading consumption evenly over time and states, optimal because U is concave.
- **Euler equation**: the FOC in C_0; marginal utility today = discounted expected marginal utility of investing the dollar.
- **Allocation rule**: the FOC in w_i; $E[U'(\tilde{C}_1)\tilde{R}_i]$ is equal across assets.
- **Gross return ($\tilde{R}$)**: what \$1 becomes, $1 + r$.
- **Hansen–Jagannathan bound [Beyond]**: the maximum Sharpe ratio is at most $\sigma(\tilde{M})/E[\tilde{M}]$.
- **Intertemporal MRS (IMRS)**: the rate at which the investor trades today's consumption for date-1 consumption in a given state, equal to $\tilde{M}$.
- **Lagrange multiplier (λ)**: the shadow value of the budget constraint $\sum w_i = 1$.
- **Marginal utility (U')**: the extra utility from one more dollar; high when consumption is low.
- **Patience (δ)**: the utility discount factor; higher means more patient.
- **Pricing kernel**: another name for the SDF; prices every payoff via $P = E[\tilde{M}\tilde{X}]$.
- **Risk-neutral probabilities [Beyond]**: $\pi_s^Q = p_s M_s R_f$; probabilities under which prices are discounted expected payoffs at R_f.
- **State price (q_s) [Beyond]**: the price today of \$1 paid only in state s; $q_s = p_s M_s$.
- **Stochastic discount factor (SDF, $\tilde{M}$)**: $\delta U'(\tilde{C}_1)/U'(C_0)$; a random discount factor, one value per state.

- **Time-separable utility**: $V = U(C_0) + \delta E[U(\tilde{C}_1)]$; each date's utility adds, with no interaction.
- **Tilde (~)**: marks a random variable.
- **Equity premium puzzle [Beyond]**: plausible risk aversion and smooth consumption give an SDF too stable to explain the observed equity Sharpe ratio.

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5. Self-test

Questions

Q1 (conceptual, easy). Explain in words why $\tilde{M}$ is high in recessions. Use this to explain why an asset that pays off only in recessions can have an expected return below the risk-free rate, and why that is not a mispricing.

Q2 (numerical, easy). Two equally likely states, with $\tilde{M} = (1.2,\ 0.7)$ in (bad, good). (a) Find the price of a riskless bond paying \$1 and R_f. (b) Price asset A with payoff (50, 150) and asset B with payoff (150, 50); find their expected gross returns and risk premia. (c) Verify each price using $P = E[\tilde{X}]/R_f + \mathrm{Cov}(\tilde{M}, \tilde{X})$.

Q3 (derivation). Starting from $\max_{C_0, w} U(C_0) + \delta E[U(\tilde{C}_1)]$ with $\tilde{C}_1 = (W_0 - C_0)\sum_i w_i \tilde{R}_i$, derive the Euler equation. Explain where the minus sign comes from and why $U'(\tilde{C}_1)$ cannot be taken outside the expectation.

Q4 (derivation). Using a Lagrangian for $\sum_i w_i = 1$, derive the allocation rule $E[U'(\tilde{C}_1)\tilde{R}_i] = E[U'(\tilde{C}_1)\tilde{R}_j]$. Then combine it with the Euler equation to obtain $E[\tilde{M}\tilde{R}_i] = 1$ and give $\tilde{M}$.

Q5 (numerical). An investor has $U = \ln C$, $\delta = 0.96$ and $W_0 = 196$. (a) Show $C_0^* = W_0/(1+\delta)$ and compute C_0^* and S^*. (b) Date-1 consumption is 90, 105 or 120 with probabilities 0.3, 0.4, 0.3. Compute $\tilde{M}$ in each state, $E[\tilde{M}]$ and R_f. (c) Price a claim paying \$1 only in the recession state and find its expected return.

Q6 (numerical). Same states as Q5 but CRRA utility with $\gamma = 2$ (keep $\delta = 0.96$, $C_0 = 100$). (a) Compute $\tilde{M}$ and R_f. (b) Price the payoff (80, 105, 130) and compute its risk premium. (c) Compare with the log case (premium 2.27%, $R_f = 8.02\%$) and explain both changes.

Q7 (derivation + numerical). (a) Derive $E[\tilde{R}_i] - R_f = -R_f \mathrm{Cov}(\tilde{M}, \tilde{R}_i)$ from $E[\tilde{M}\tilde{R}_i] = 1$. (b) Using Q2's data, compute $\sigma(\tilde{M})/E[\tilde{M}]$ and asset A's Sharpe ratio. Why is the bound attained exactly here? (c) A fund in this economy claims a Sharpe ratio of 0.40. What do you conclude?

Q8 (conceptual, trap-spotting). For each statement, say true/false and correct if false. (a) "Assets with higher return variance must have higher expected returns." (b) "$P = E[\tilde{M}]E[\tilde{X}]$." (c) "Using net returns, $E[\tilde{M}\tilde{r}] = 1$." (d) "$\delta$ is the discount factor applied to dollars." (e) "The Lagrange multiplier λ in [S13] measures risk aversion."

Q9 (derivation, hard). Suppose $\tilde{M} = a - b\tilde{R}_M$ with $b > 0$. Derive the CAPM. Explain why $b > 0$ implies that negative-β assets earn less than R_f, linking to W3.

Q10 (hard, synthesis). (a) Show that with discrete states, $P = \sum_s q_s X_s$ with $q_s = p_s M_s$, and that $\pi_s^Q = q_s R_f$ are probabilities. (b) Compute q and π^Q for Q5's economy and compare with p. (c) Explain in one paragraph why "every asset pricing model is a model of $\tilde{M}$", naming what $\tilde{M}$ is in the CAPM, the W4 factor model and this week's consumption model.

Solutions

S1.

- $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$ and $U'' < 0$, so U' is high when $\tilde{C}_1$ is low. In a recession consumption is low, so an extra dollar there is worth a lot of utility, and $\tilde{M}$ is high [S16].

- A recession-only payoff is concentrated where $\tilde{M}$ is high, so $P = E[\tilde{M}\tilde{X}]$ is high and its expected return $E[\tilde{X}]/P$ is low, possibly below R_f.
- Holding it is rational: it is **insurance**. The low expected return is the premium paid for a payoff that arrives "exactly when a dollar is worth most". It is correctly priced; its $\text{Cov}(\tilde{M}, \tilde{R}) > 0$ justifies a negative premium.

S2.

- (a) $P_{bond} = E[\tilde{M}] = 0.5(1.2) + 0.5(0.7) = \mathbf{0.95}$, so $R_f = 1/0.95 = \mathbf{1.0526}$ (5.26%).
- (b) A: $P = 0.5(1.2)(50) + 0.5(0.7)(150) = 30 + 52.5 = \mathbf{82.5}$; $E[\tilde{X}] = 100$, so $E[\tilde{R}] = 100/82.5 = \mathbf{1.2121}$, premium **+15.9%**.
- B: $P = 0.5(1.2)(150) + 0.5(0.7)(50) = 90 + 17.5 = \mathbf{107.5}$; $E[\tilde{R}] = 100/107.5 = \mathbf{0.9302}$, premium **−12.2%** (below R_f and below zero: B is insurance).
- (c) $\text{Cov}(\tilde{M}, \tilde{X}_A) = E[\tilde{M}\tilde{X}_A] - E[\tilde{M}]E[\tilde{X}_A] = 82.5 - 95 = -12.5$, so $100/1.0526 - 12.5 = 95 - 12.5 = 82.5$ ✓. For B: Cov $= 107.5 - 95 = +12.5$, so $95 + 12.5 = 107.5$ ✓.

S3.

- $dV/dC_0 = U'(C_0) + \delta E[U'(\tilde{C}_1)\, \partial \tilde{C}_1/\partial C_0] = 0$.
- $\partial \tilde{C}_1/\partial C_0 = -\sum_i w_i^* \tilde{R}_i$: consuming \$1 more today removes \$1 from the invested pot, and that dollar would have grown at $\tilde{R}_p$. Hence the minus sign.
- Result: $U'(C_0^*) = \delta E[U'(\tilde{C}_1^*) \sum_i w_i^* \tilde{R}_i]$.
- $U'(\tilde{C}_1)$ is random and depends on $\tilde{R}_p$ itself (through $\tilde{C}_1$), so $E[U'\tilde{R}_p] = E[U']E[\tilde{R}_p] + \text{Cov}(U', \tilde{R}_p)$; the covariance term is exactly the risk adjustment.

S4.

- $\mathcal{L} = \delta E[U(\tilde{C}_1)] - \lambda(\sum_j w_j - 1)$; $\partial \tilde{C}_1/\partial w_i = (W_0 - C_0)\tilde{R}_i$.
- FOC: $\delta(W_0 - C_0)E[U'(\tilde{C}_1)\tilde{R}_i] - \lambda = 0$ for all i. The left side must equal the same λ for every i, so $E[U'\tilde{R}_i] = E[U'\tilde{R}_j]$.
- Euler: $U'(C_0) = \delta \sum_i w_i^* E[U'\tilde{R}_i]$. All terms are equal and $\sum w_i^* = 1$, so $U'(C_0) = \delta E[U'\tilde{R}_i]$ for each i.
- Divide by $U'(C_0)$: $E[\tilde{M}\tilde{R}_i] = 1$ with $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$. ■

S5.

- (a) $1/C_0 = \delta E[\tilde{R}_p/((W_0 - C_0)\tilde{R}_p)] = \delta/(W_0 - C_0)$, so $W_0 - C_0 = \delta C_0$ and $C_0^* = W_0/(1+\delta) = 196/1.96 = \mathbf{100}$; $S^* = \mathbf{96}$.
- (b) $M_s = 0.96 \times 100/C_{1,s}$: **(1.0667, 0.9143, 0.8000)**. $E[\tilde{M}] = 0.3200 + 0.3657 + 0.2400 = \mathbf{0.9257}$, so $R_f = \mathbf{1.0802}$ (8.02%).
- (c) $P = 0.3 \times 1.0667 \times 1 = \mathbf{0.320}$; $E[\tilde{X}] = 0.3$, so $E[\tilde{R}] = 0.3/0.32 = \mathbf{0.9375}$: an expected loss of 6.25%, a premium of **−14.3%** vs R_f.

S6.

- Here consumption in each state is taken as given (as in an endowment economy); we only use $\tilde{M}$ to price payoffs, so no re-optimisation of C_0 is needed.
- (a) $M_s = 0.96(C_{1,s}/100)^{-2}$: $0.96/0.81 = \mathbf{1.1852}$, $0.96/1.1025 = \mathbf{0.8707}$, $0.96/1.44 = \mathbf{0.6667}$. $E[\tilde{M}] = 0.3556 + 0.3483 + 0.2000 = 0.9039$, so $R_f = \mathbf{1.1064}$ (10.6%).
- (b) $P = 0.3(1.1852)(80) + 0.4(0.8707)(105) + 0.3(0.6667)(130) = 28.44 + 36.57 + 26.00 = \mathbf{91.02}$. $E[\tilde{R}] = 105/91.02 = \mathbf{1.1536}$, premium **+4.73%**.
- (c) Both rise vs log utility.
  - **Premium** ↑: higher γ makes U' steeper, so $\tilde{M}$ is more dispersed and covaries more strongly (negatively) with the pro-cyclical payoff.
  - R_f ↑: consumption is expected to grow ($E[\tilde{C}_1] = 105 > 100$); a more curved U makes the investor want to smooth by borrowing against the richer future, and a higher R_f is needed to stop them.

S7.

- (a) $1 = E[\tilde{M}\tilde{R}_i] = E[\tilde{M}]E[\tilde{R}_i] + \mathrm{Cov}(\tilde{M}, \tilde{R}_i)$. Divide by $E[\tilde{M}] = 1/R_f$: $E[\tilde{R}_i] = R_f - R_f \mathrm{Cov}(\tilde{M}, \tilde{R}_i)$. ■
- (b) $\sigma(\tilde{M}) = 0.25$ and $E[\tilde{M}] = 0.95$, so the bound is **0.2632**. Asset A: $\tilde{R}_A = (0.6061, 1.8182)$, $E = 1.2121$, $\sigma = 0.6061$, so Sharpe $= (1.2121 - 1.0526)/0.6061 =$ **0.2632**. It is on the bound because with two states every payoff is perfectly (negatively, here) correlated with $\tilde{M}$: $|\mathrm{Corr}| = 1$.
- (c) $0.40 > 0.263$ violates the HJ bound: in this economy no asset or portfolio can have that Sharpe ratio. Either the claim is wrong (estimation noise, hidden risks, look-ahead) or $\tilde{M}$ is not what we assumed.

S8.

- (a) **False.** Only covariance with $\tilde{M}$ is priced. A high-variance payoff with $\mathrm{Cov}(\tilde{M}, \tilde{R}) = 0$ earns R_f.
- (b) **False.** $P = E[\tilde{M}\tilde{X}] = E[\tilde{M}]E[\tilde{X}] + \mathrm{Cov}(\tilde{M}, \tilde{X})$; the equality holds only when the covariance is zero.
- (c) **False.** Gross: $E[\tilde{M}(1 + \tilde{r})] = 1$, so $E[\tilde{M}\tilde{r}] = 1 - E[\tilde{M}] = 1 - 1/R_f$.
- (d) **False.** δ discounts utility. The dollar discount factor is $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$, and on average $E[\tilde{M}] = 1/R_f$.
- (e) **False.** It is the shadow value of the constraint $\sum w_i = 1$, equal to $(W_0 - C_0)U'(C_0^*)$ at the optimum.

S9.

- $R_f = 1/E[\tilde{M}]$. From the covariance form: $E[\tilde{R}_i] - R_f = -R_f \mathrm{Cov}(a - b\tilde{R}_M, \tilde{R}_i) = R_f b \,\mathrm{Cov}(\tilde{R}_M, \tilde{R}_i)$.
- Apply to the market itself: $E[\tilde{R}_M] - R_f = R_f b \,\mathrm{Var}(\tilde{R}_M)$, so $R_f b = (E[\tilde{R}_M] - R_f)/\mathrm{Var}(\tilde{R}_M)$.
- Substitute: $E[\tilde{R}_i] - R_f = \dfrac{\mathrm{Cov}(\tilde{R}_i, \tilde{R}_M)}{\mathrm{Var}(\tilde{R}_M)}(E[\tilde{R}_M] - R_f) = \beta_i (E[\tilde{R}_M] - R_f)$. ■
- $b > 0 \Rightarrow \tilde{M}$ is high when $\tilde{R}_M$ is low (market crashes are the bad states). An asset with $\beta < 0$ pays when the market falls, so $\mathrm{Cov}(\tilde{M}, \tilde{R}_i) > 0$ and its premium is negative: W3's "insurance" result [W3 S42].

S10.

- (a) $P = E[\tilde{M}\tilde{X}] = \sum_s p_s M_s X_s = \sum_s q_s X_s$. $\pi_s^Q = q_s R_f \geq 0$ (since $M_s > 0$ when $U' > 0$) and $\sum_s \pi_s^Q = R_f \sum_s p_s M_s = R_f E[\tilde{M}] = 1$. So $P = \sum_s \pi_s^Q X_s / R_f = E^Q[\tilde{X}]/R_f$.
- (b) $q = (0.3 \times 1.0667,\ 0.4 \times 0.9143,\ 0.3 \times 0.8) =$ **(0.320, 0.366, 0.240)**; $\pi^Q = q \times 1.0802 =$ **(0.346, 0.395, 0.259)** vs $p = (0.30, 0.40, 0.30)$. The risk-neutral measure puts **more** weight on the recession and **less** on the boom.
- (c) Every model says $P = E[\tilde{M}\tilde{X}]$; they differ only in what $\tilde{M}$ is and what it depends on.
  - CAPM: $\tilde{M} = a - b\tilde{R}_M$ (bad states = market crashes).
  - W4 multi-factor: $\tilde{M} = a - b^\top \tilde{f}$ (bad states = bad factor realisations).
  - This week: $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$ (bad states = low consumption).
  - Testing a model = testing whether its $\tilde{M}$ prices all assets, $E[\tilde{M}\tilde{R}_i] = 1$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title: Stochastic Discount Factor	header
S2	"How do you spend your salary?"	1.1
S3	Spend (C_0) or invest ($S = W_0 - C_0$)	1.1
S4	Uncertainty; tilde = random; $\tilde{C}_1 = (W_0 - C_0)\tilde{R}$	1.2
S5	"A dollar is worth more when you are poor" ⚠ typo	1.3
S6	Diminishing marginal utility; $U' > 0$, $U'' < 0$; smooth consumption	1.3
S7	One period, two dates; two decisions; symbol table	1.4
S8	Budget: $\sum w_i = 1$, $\tilde{C}_1 = (W_0 - C_0)\sum w_i \tilde{R}_i$; gross returns	1.4
S9	Objective $U(C_0) + \delta E[U(\tilde{C}_1)]$	1.5
S10	Time-separable utility V	1.5
S11	Problem, Lagrangian, two FOCs ⚠ typesetting	1.6
S12	FOC in C_0: Euler equation	1.7
S13	FOC in w_i: allocation rule, λ	1.8
S14	Combine: $E[\tilde{M}\tilde{R}_i] = 1$	1.9
S15	SDF definition, names, $P_i = E[\tilde{M}\tilde{X}_i]$ ⚠ units of the IMRS	1.10
S16	SDF largest in bad states; insurance dear, boom assets cheap	1.11
S17	In-class quiz Q1–3 (title only)	1.13

Appendix B. Flag summary (⚠)

1. **[S12]-[S15]** The deck's internal slide numbers ("slide 9, 10, 11") are 3 lower than PDF pages ([S12], [S13], [S14]).
2. **[S5]** "Consuming 1 dollar is worth more you are poor": missing "when".
3. **[S11] vs [S13]** Lagrangian written $+\lambda(1 - \sum w_i)$ vs $-\lambda(\sum w_j - 1)$: identical. [S13] also drops $U(C_0)$ from the weight problem, which is harmless.
4. **[S13]** λ is a Lagrange multiplier (shadow value of the budget = $(W_0 - C_0)U'(C_0^*)$), not risk aversion (W2/W5) or a factor premium (W5).
5. **[S4] vs [S8]** $\tilde{R}$ must be the **gross** return (stated only on [S8]). With net returns, $E[\tilde{M}\tilde{r}] = 1 - E[\tilde{M}]$, not 1.
6. **[S9]-[S10]** δ discounts utility, not dollars. The dollar discount factor is $\tilde{M}$.
7. **[S14]** "Holds for every asset" requires every weight to be interior (no binding short-sale or position limits); an asset stuck at a constraint satisfies only an inequality.
8. **[S16]** M_s is the value today of \$1 in state s **per unit probability**; the actual price of that claim is the state price $q_s = p_s M_s$.
9. **[S17]** Quiz slide shows a title only; no questions in the deck.
10. **[S15]** The IMRS box says a future dollar is worth "units of today's happiness". $\tilde{M}$ is in today's **dollars** (utility ÷ marginal utility), which is why $P = E[\tilde{M}\tilde{X}]$ is a price.
11. **[S11]** The objective is typeset "max C_0, w" (a literal underscore); read it as $\max_{C_0, w}$. The slide lists only $\sum_i w_i = 1$ as the constraint; the budget $\tilde{C}_1 = (W_0 - C_0)\sum_i w_i \tilde{R}_i$ from [S8] is substituted into the objective.